



Syntax III

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2026/09/30 (updated: 2026-09-27)

Please pick up your **name card** from the front

Previously on 201: Constituent and Its Types

- **constituent**: a group of words that **acts together** as a **unit**
- **phrases**: types of constituents, different types are decided by the **part of speech** of the **head**
 - **noun phrase (NP)**: man, the man, the happy man, the man with the hat
 - **prepositional phrase (PP)**: into the forest, by the author, for good luck
 - **verb phrase (VP)**: eats, eats pizza, eats quickly, eats pizza quickly at noon
 - **adjective phrase (AdjP)**: happy, ugly
 - **adverb phrase (AdvP)**: quickly, happily
- **constituency tests**: to test whether a string of words is a constituent or not
 - replacement, movement, fragment, coordination

I saw the man with a telescope.

this sentence has **two** possible interpretations

what are they?

in each interpretation, who is *with a telescope*

I saw the man with a telescope.

Interpretation 1

I used a telescope and I saw a man.



Interpretation 2

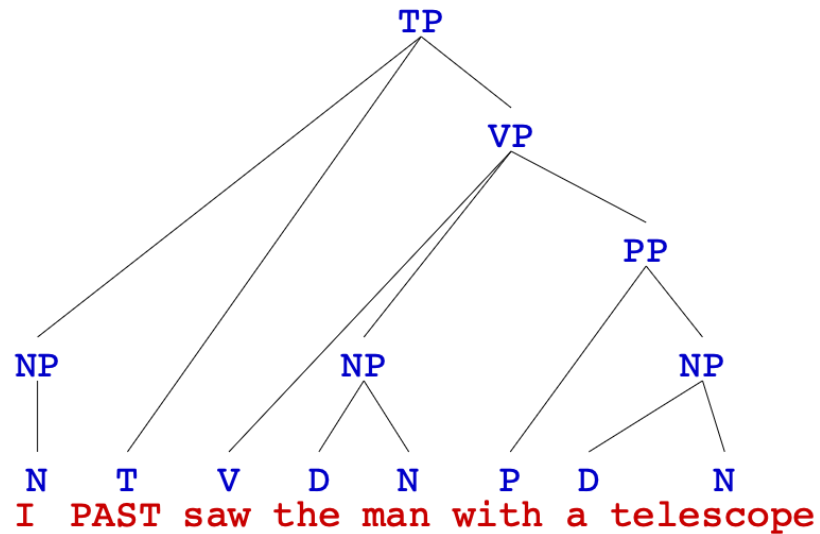
I saw the man who has a telescope.



I saw the man with a telescope.

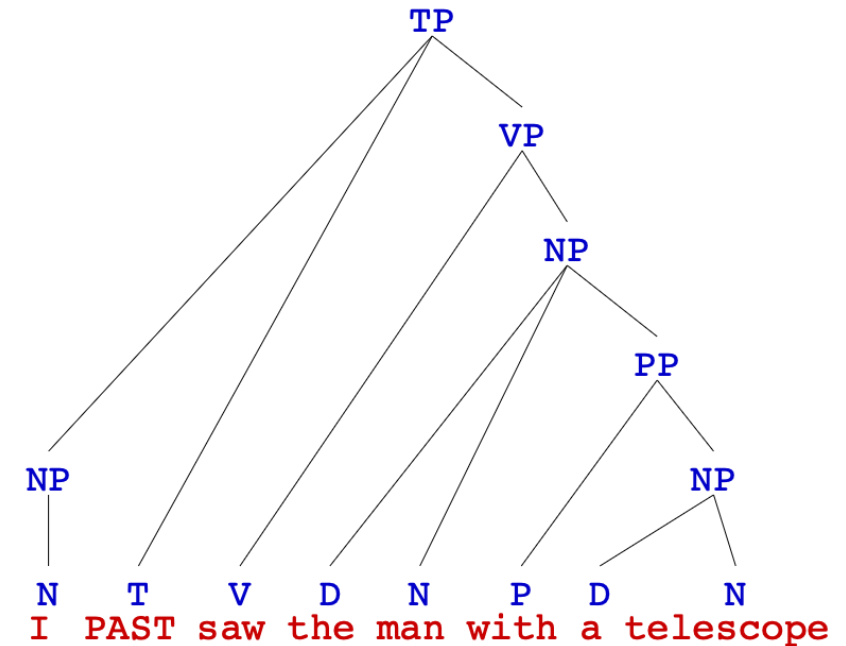
Interpretation 1

I used a telescope and I saw a man.



Interpretation 2

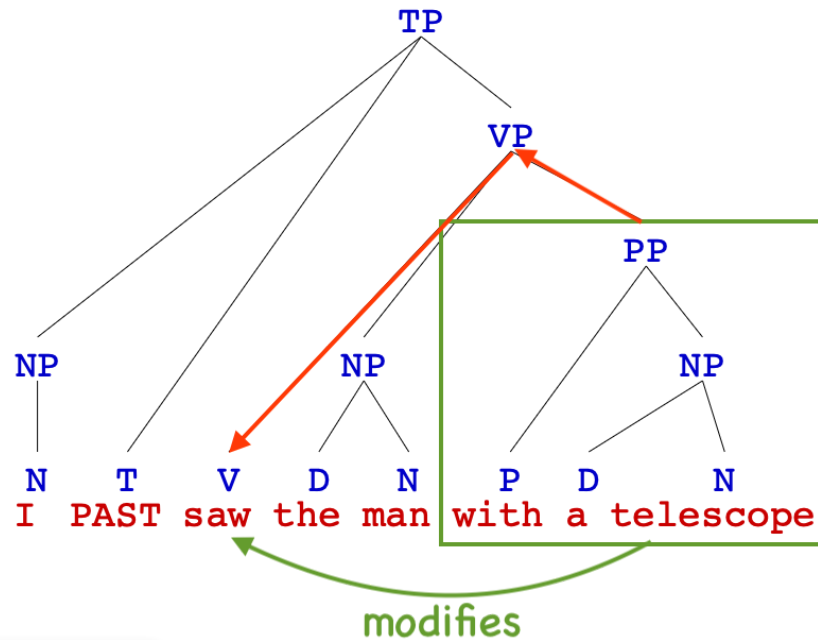
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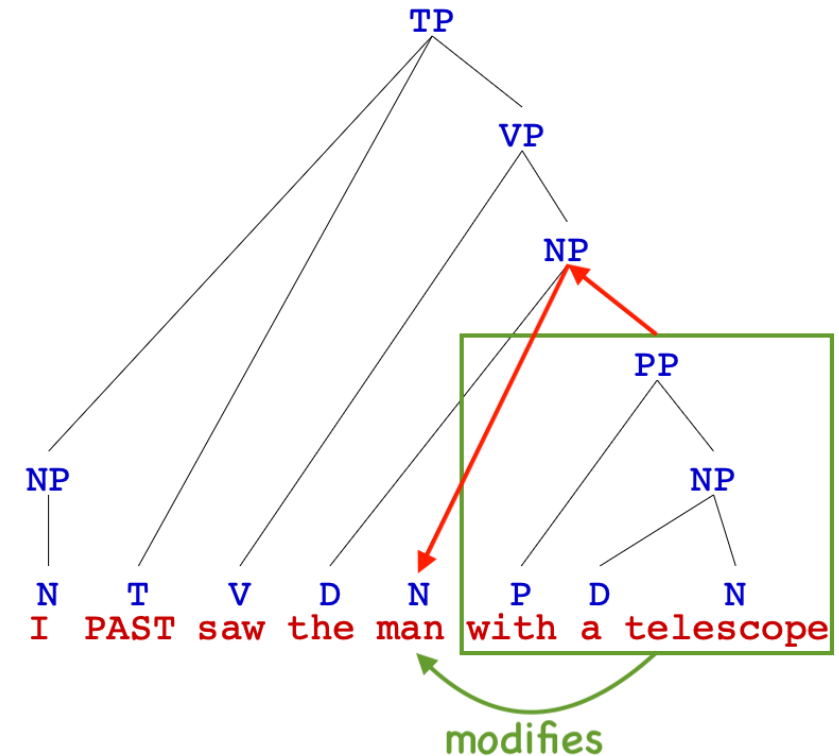
Interpretation 1

I used a telescope and I saw a man.



Interpretation 2

I saw the man who has a telescope.



I saw the man with a telescope.

an example of **structural ambiguity**

but we did not look at it via **syntax trees** to visualize its two possible structures

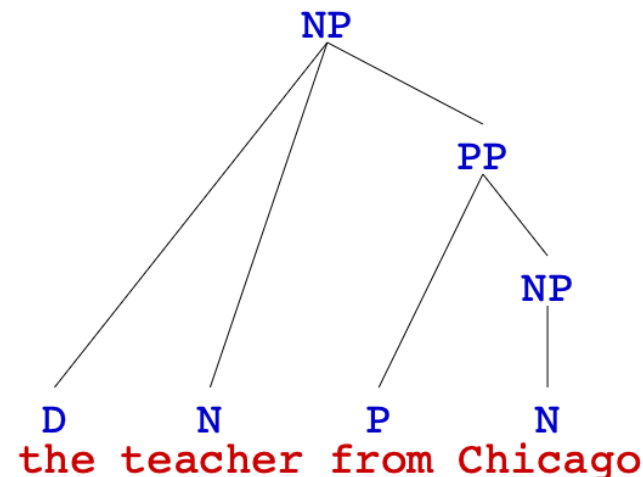
the syntax trees on the previous page show how there are two ways to interpret what [with a telescope]. **modifies**

the meaning of a sentence $\neq$ sum of its words, **syntactic structure** also contributes to meaning

Phrase Structure Rules (PSRs)

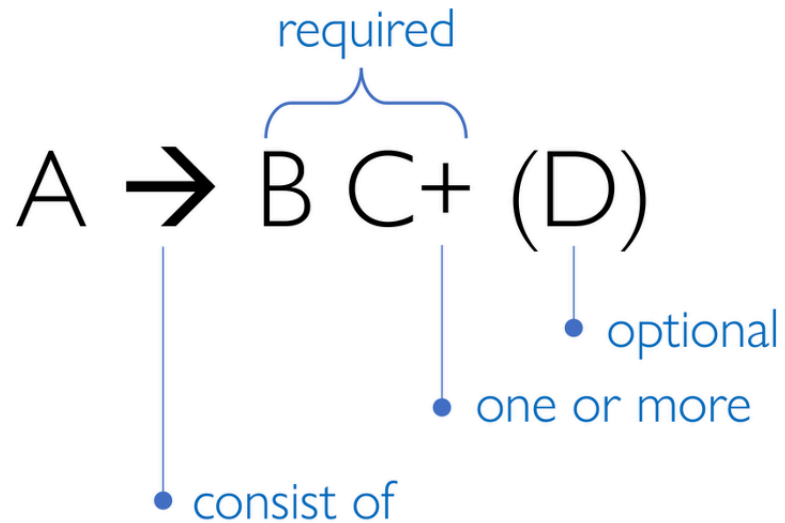
- there are (descriptive) rules that indicate **how phrases are built out of smaller units** (which could be words or other phrases)
 - these rules are called **phrase structure rules**
 - they indicate the order and position of the smaller units in a phrase
- we derive these rules by observing what are the possible structures that exist in **natural language**

NP → (D) (AdjP+) **N** (PP+)

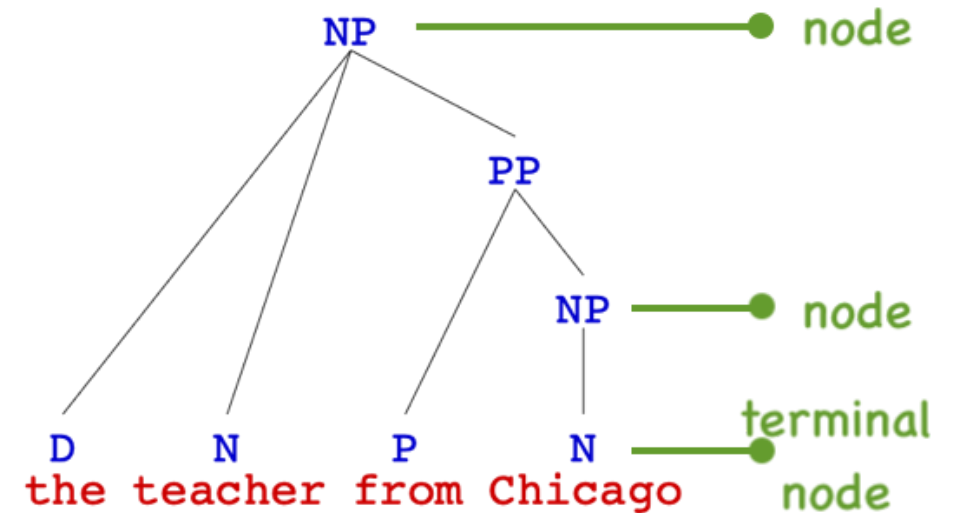


Notational Notes

phrase structure rules indicate the **order** and **position** of the sub-parts



NP $\rightarrow$ (D) (AdjP+) **N** (PP+)



Phrase Structure Rules for English

- **NP** $\rightarrow$ (D) (AdjP+) **N** (PP+)
- **TP** $\rightarrow$ NP **(T)** VP
- **PP** $\rightarrow$ **P** NP
- **CP** $\rightarrow$ **C** TP
- **AdjP** $\rightarrow$ (AdvP) **Adj**
- **AdvP** $\rightarrow$ (AdvP) **Adv**
- **VP** $\rightarrow$ (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)

Noun Phrases (NP)

- a noun phrase consists of **a noun** and all the words that **modify** it
- the simplest NP is a single noun:
 - *Harriet, dogs, soup, snow*
- an **NP** can also include any of the following:
 - a **determiner**
the hat, *my* shoes, *this* idea, *which* tree
 - one or more **adjective phrases** (AdjP)
blue sky, a *great* time, *big red* trucks
 - one or more prepositional phrases (PP)
 spaghetti *with meatballs*, trees *in the yard*, the tall bridge *over the river*

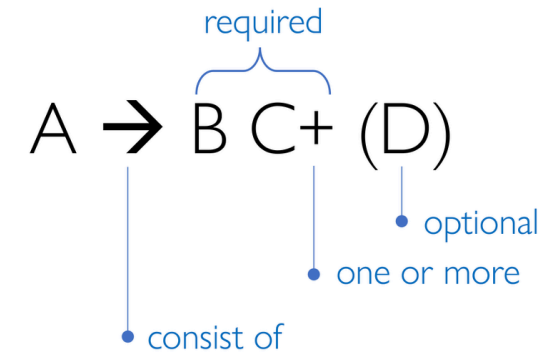
- **note:** an adjective will **always** go in a separate AdjP, for reasons we'll discuss soon in a few pages later

CONCORD WITHIN A NOUN PHRASE:

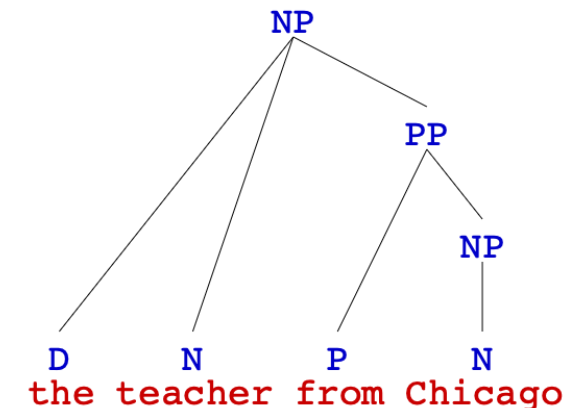


Phrase Structure Rules: Take NP as an Example

- **notational notes** once again:
 - order **matters** – elements always appear in the order given in the rule
 - parentheses = **optional** element
 - no parentheses = **obligatory** element
 - plus sign (+) = one or more
- so the rule for **noun phrases** means that a noun phrase consists of (in order):
 - optional determiner
 - one or more optional adjective phrase(s)
 - the **noun**
 - one or more optional prepositional



NP $\rightarrow$ (D) (AdjP⁺) **N** (PP⁺)



Syntax Trees Drawn by Phrase Structure Rules

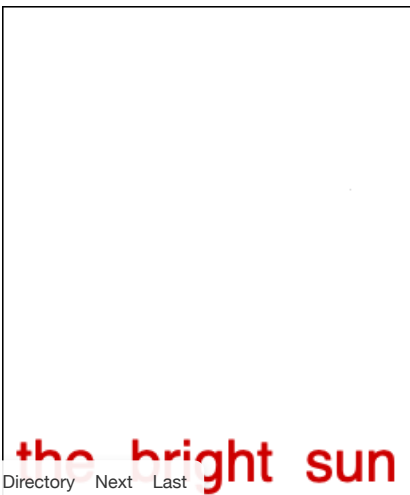
1. write the **words** at the bottom
2. label their **parts of speech**
3. figure out how they **combine** into phrases
4. write phrase type above and connect **all** its elements with lines – some phrases contain other phrases
5. tree should reflect an **existing PSR**: phrase type on left of PSR at top elements on right of PSR at bottom

the bright sun

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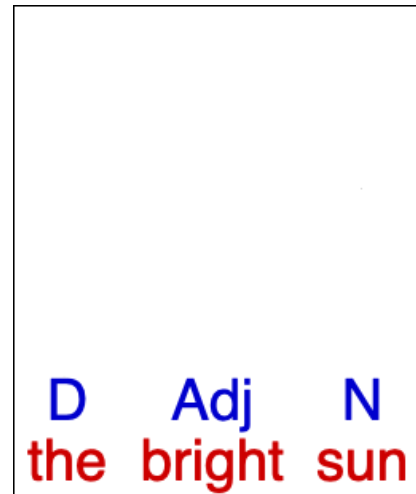
the bright sun



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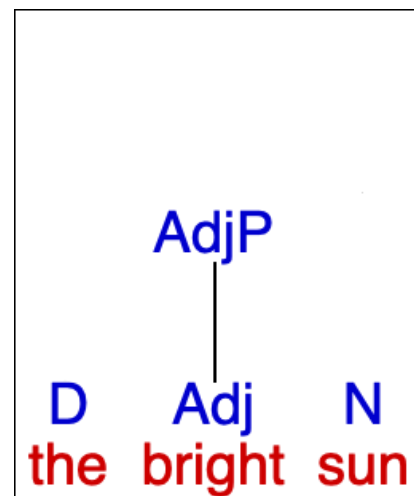
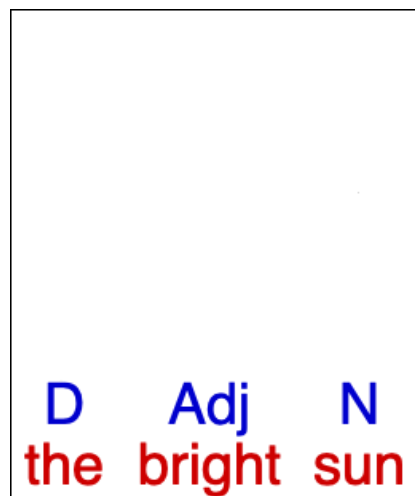
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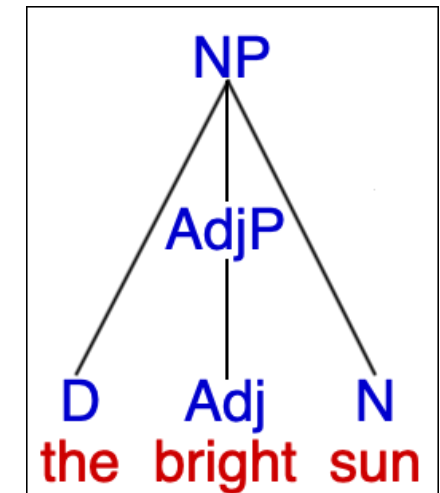
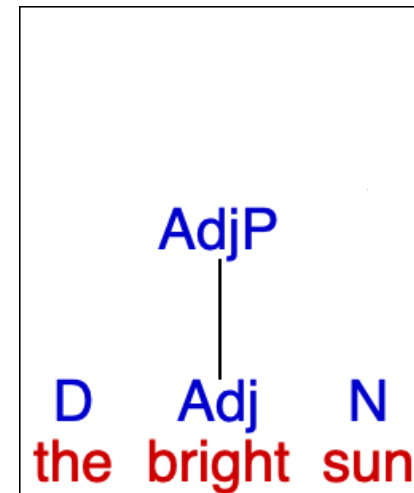
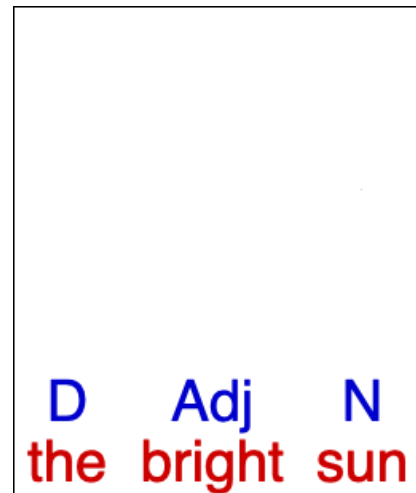
the bright sun



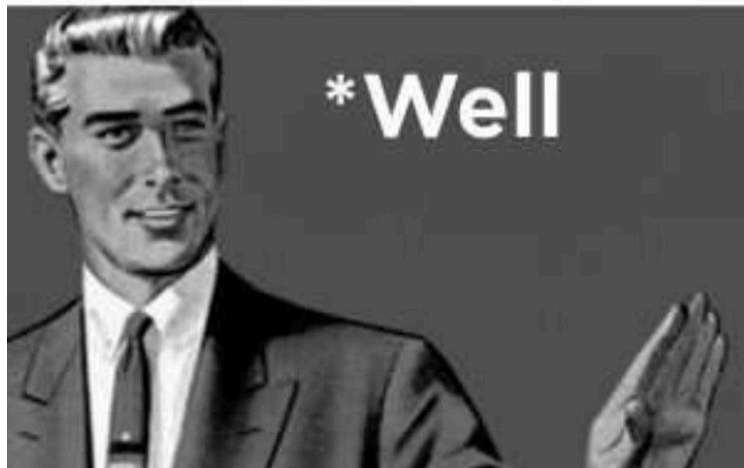
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the bright sun



Adjective Phrase (AdjP), Adverb Phrase (AdvP)



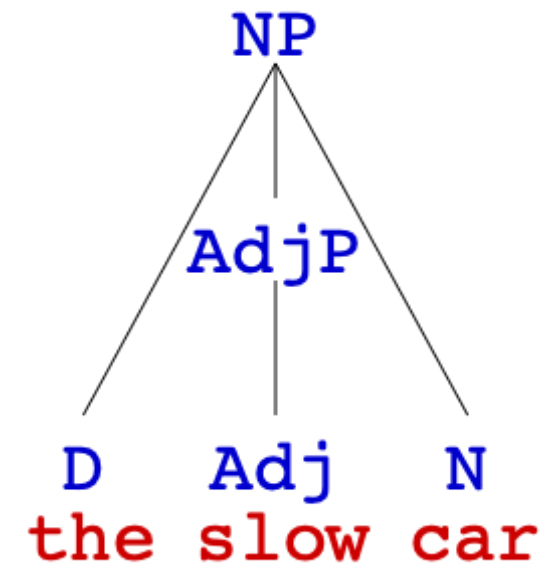
consider the following sentence, and think about whether *happy* and *quickly* are constituents?

The happy dog ran quickly.

- **constituency tests:**
 - How was the dog who ran quickly? *Happy!*
(stand alone)
 - How did the happy dog run? *Quickly!*
(stand alone)
 - The happy dog ran like that. (replacing *quickly*)
- **adjectives** and **adverbs** are their own constituents, so they need to be put into their own phrases: **AdjP** and **AdvP**

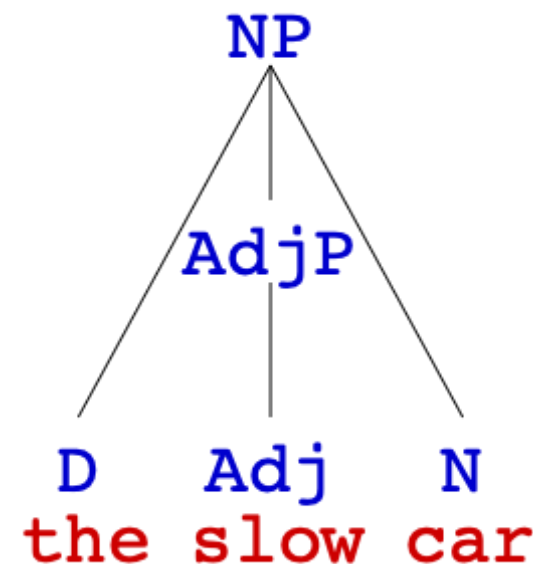
Adjective Phrases (AdjP)

- most basic **AdjP**: one adjective (AdjP → Adj)
 - it can be inserted into an **NP** as shown on the right
- **Modification Rule**
 - *a phrase that modifies a word must be a **sibling** of that word*
 - make a guess, what is this sibling?



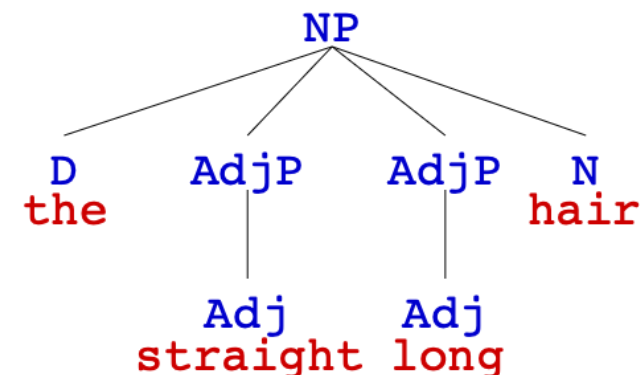
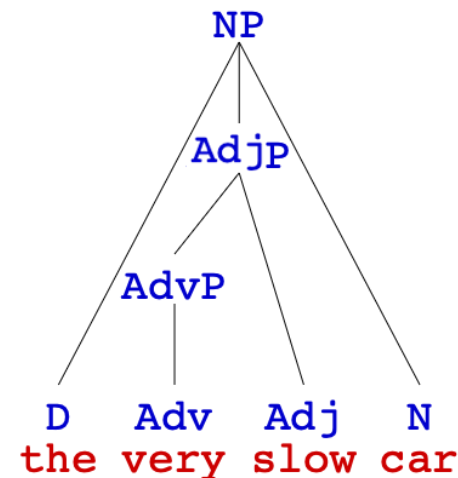
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 - *a phrase that modifies a word must be a **sibling** of that word*
 - make a guess, what is this sibling?
- by making the **AdjP** for slow a sibling of the **N** car, we're showing that slow modifies car
- we can verify they are siblings by checking they have the same **parent**: both the **AdjP** and the **N** car have **NP** as parent

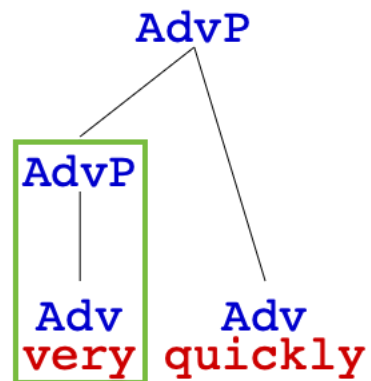
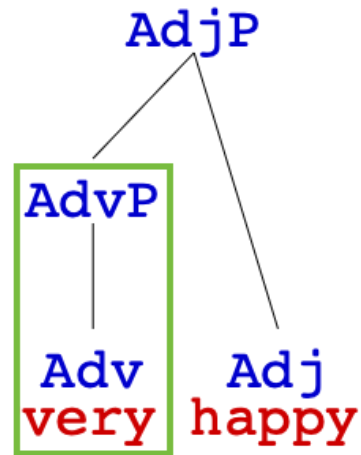


Adjective Phrases (AdjP)

- **AdjPs** can optionally contain an **adverb phrase (AdvP)**: very slow
- PSR for **AdjP** modified
AdjP → **(AdvP) Adj**
- since the adverb modifies the adjective:
 - the adverb needs to go in its own **AdvP**
 - that **AdvP** needs to be a sibling of the adjective
 - that is, they need to be **children** of the same **AdjP**
- when there are **two** adjectives in one phrase:
 - they each is a separate **AdjP**
 - since both of them are modifying the **noun**



Adverb Phrases (AdvP)



if we check the PSR rules for English **adverb phrase (AdvP)**, we will notice that they are basically the same:

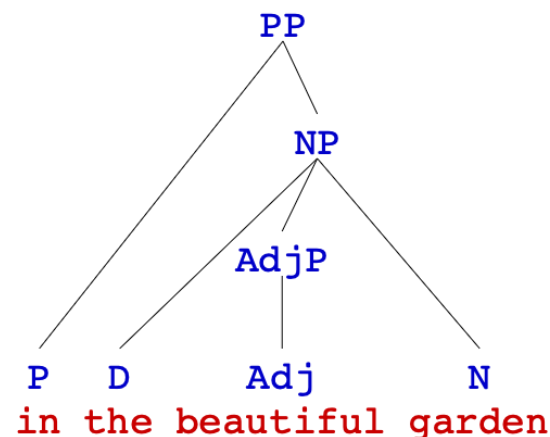
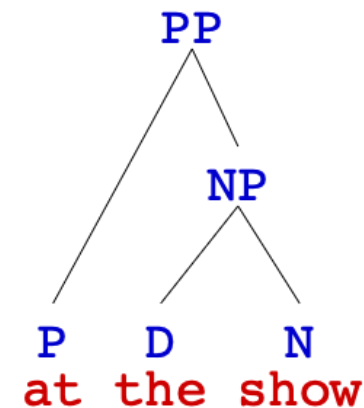
- **AdjP** → (AdvP) **Adj**
- **AdvP** → (AdvP) **Adv**
- if you have two adverbs in a row, they will be in nested structure
 - the first one **modifies** the second

Prepositional Phrases (PP)

- the PSR for **PP** is very straightforward
- it is always a **PP** containing the **head** P and an **NP**

PP → **P** **NP**

- other examples
 - in** the park*
 - by** the tree*
 - with** a friend*
 - under** the sky*
 - for** my mom*
 - ...

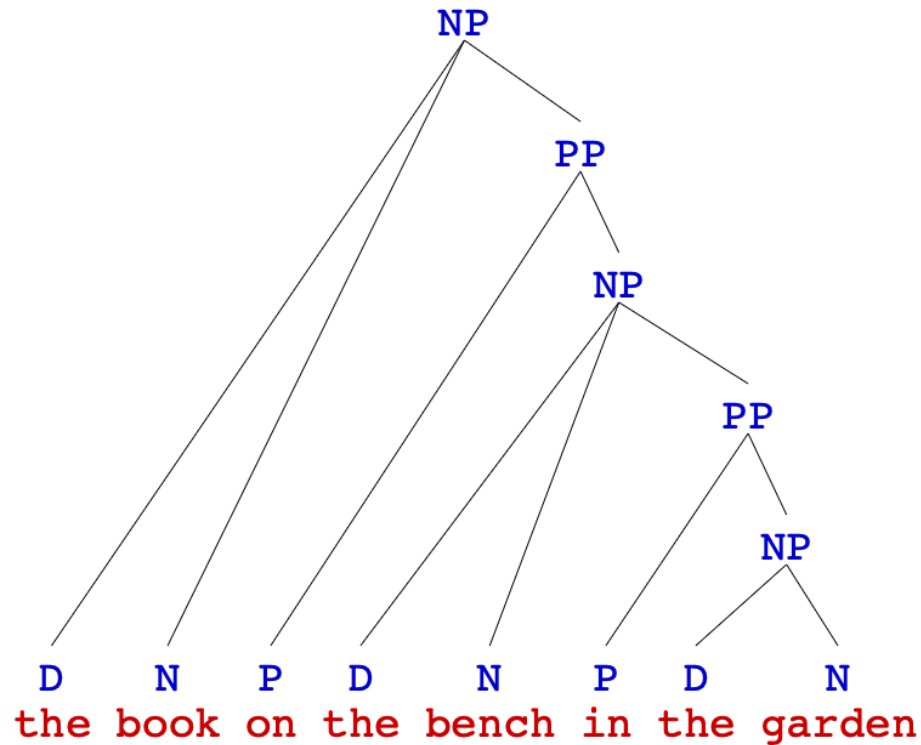


Prepositional Phrases (PP)

what's **noticeable** if we put these two PSRs together

$$\text{PP} \rightarrow \text{P NP}$$

$$\text{NP} \rightarrow (\text{D}) (\text{AdjP}+) \text{N} (\text{PP}+)$$

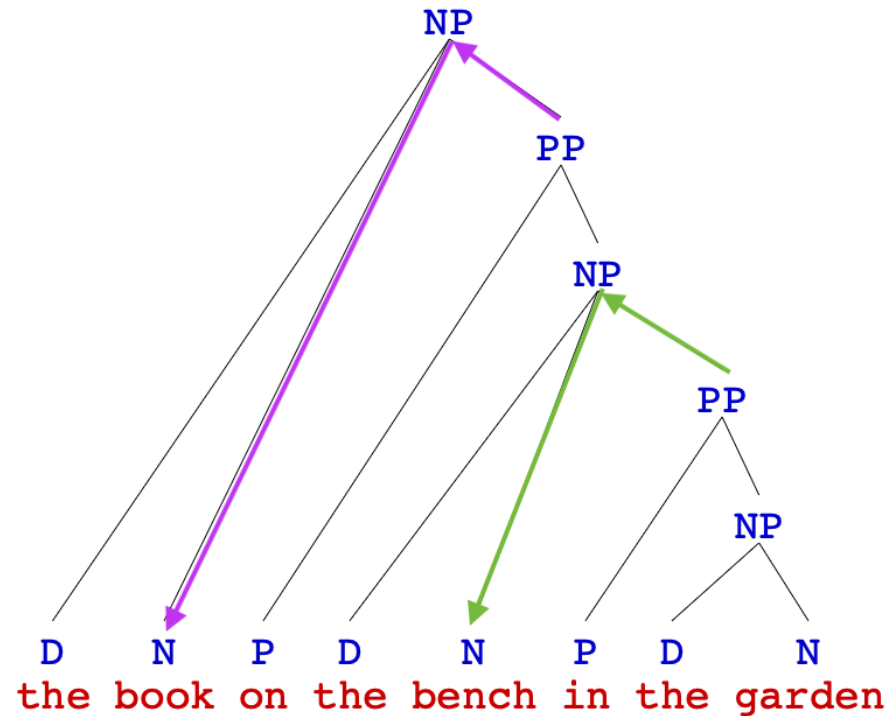


- **PP** can contain **NP**, but **NP** can contain **PP** too
- so we can have **NP** and **PP** nested to each other, go on and go on forever
- this is an example of **recursion**, which is an important feature of human language

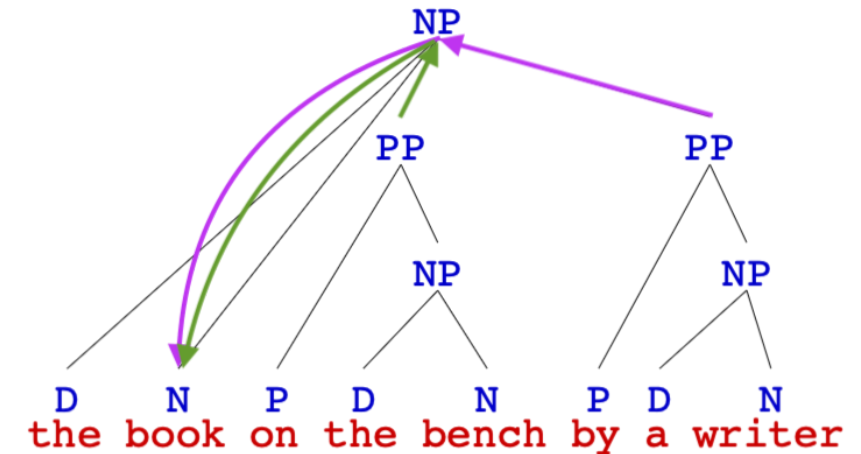
Prepositional Phrases (PP)

these phrases both contain **two PPs**, but have **different** syntactic structures, why?

- recall the **modification rule** – a phrase that modifies a word must be its **sibling**



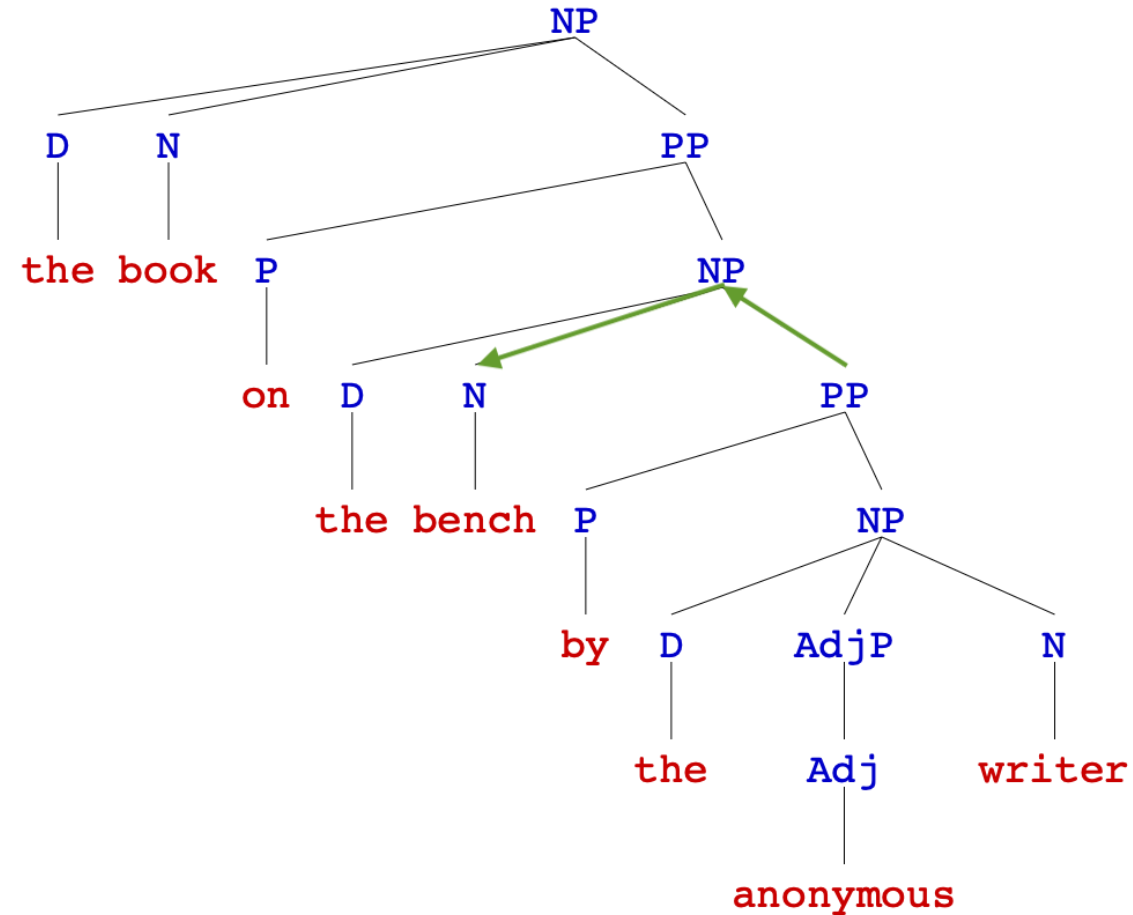
the book on the bench in the garden



the book on the bench by a writer

Prepositional Phrases (PP)

if we have this tree, what does it mean?



Exercise I

NP, **AdjP**, **AdvP**, **PP**

draw a **syntax tree** for the underlined phrases

- I ate an apple.
- John went into the building.
- They're like peas in a pod.
- The bakery sells wonderfully fresh bread.
- I like to drive really slowly.
- Mary helped the man in the car with the flat tire.
- I threw a ball to the dog with a collar in the park.

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

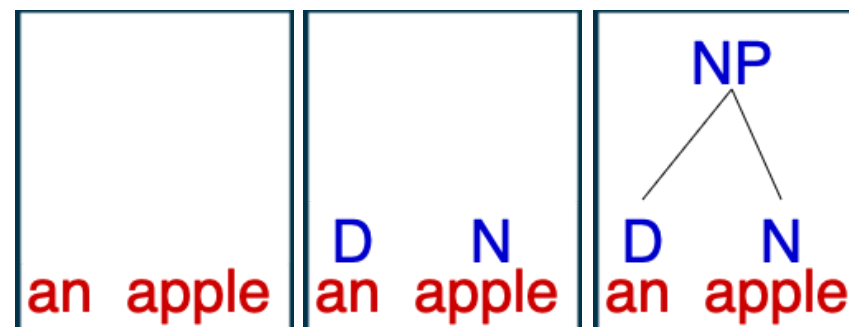
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)



- in **f** does with the flat tire modify 'the man' or 'the car'?
- what about in the park in **g**?

Exercise I

NP, AdjP, AdvP, PP

NP → (D) (AdjP+) N (PP+)

PP → P NP

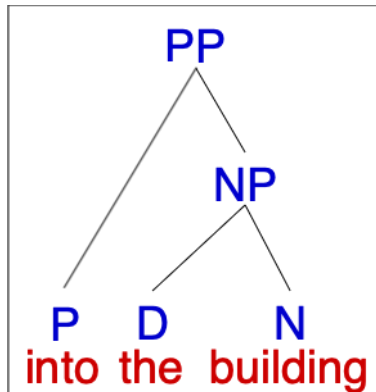
AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

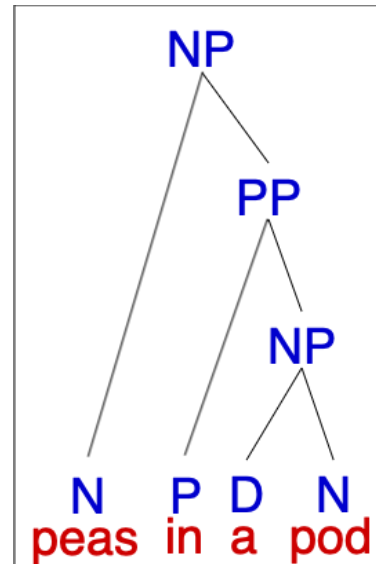
TP → NP (T) VP

CP → C TP

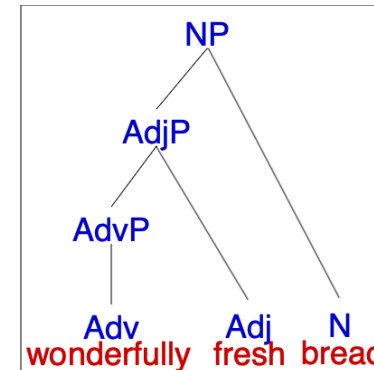
VP → (AdvP+) V (NP) (NP) (AdvP+) (PP+) (AdvP+)



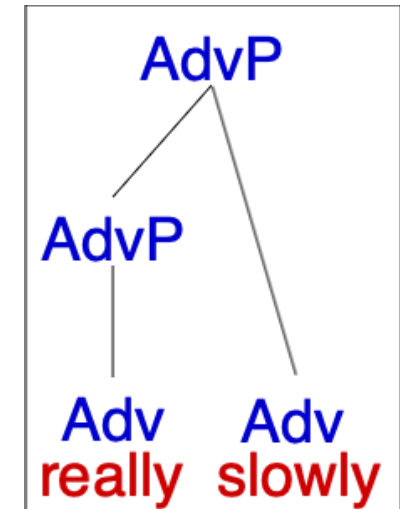
b. 'into the building'



c. 'peas in a pod'



d. 'wonderfully fresh bread'

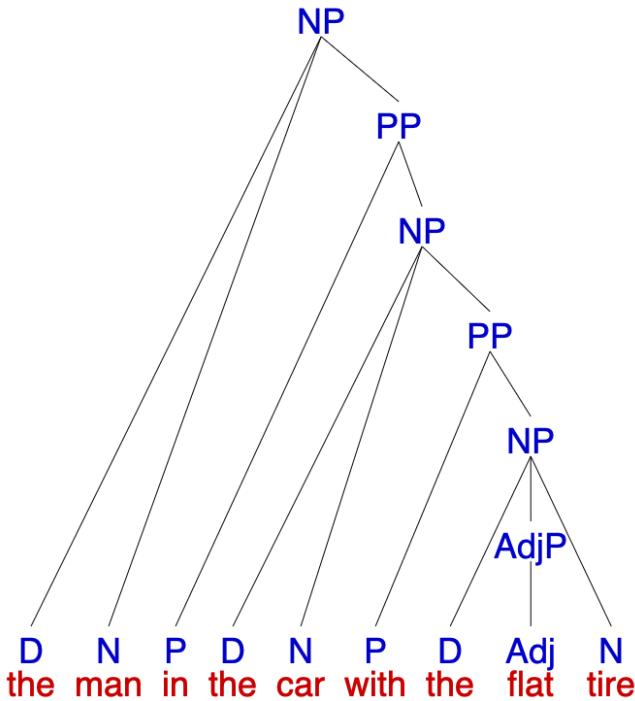


e. 'really slowly'

Exercise I

NP, AdjP, AdvP, PP

f. the man in the car with the flat tire



NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

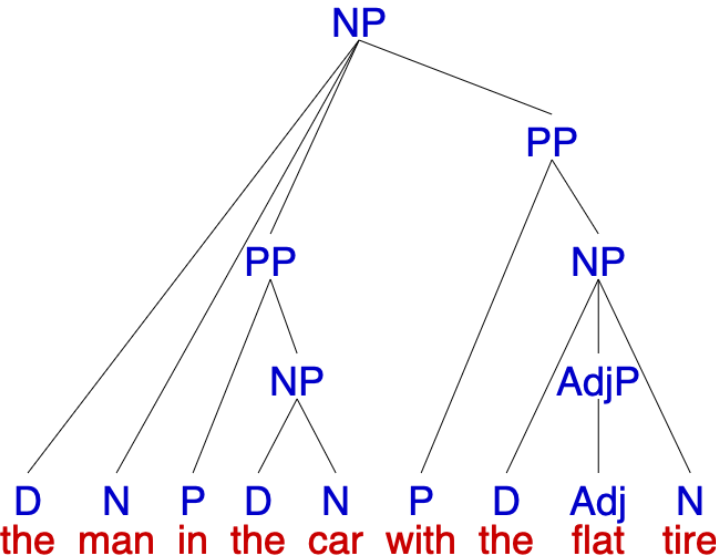
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

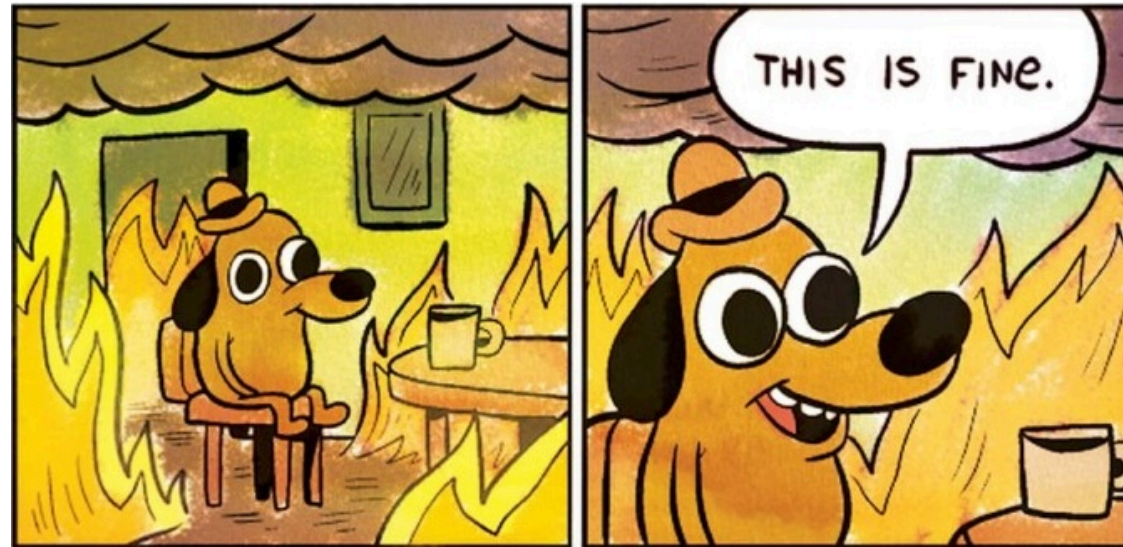
VP → (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)



Homework II is due this Friday (**Oct 02**)

Homework III will be published today (**Sept 30**) which is due next Friday (**Oct 09**)

continue reading Chapter 3: Constituency, Trees, and Rules from *Andrew Carnie's Syntax textbook*



Slides created via the R package **xaringan**.